

Quantum Algorithms

Lecture 1

Introduction to quantum mechanics and computation

Overview

- ▶ Linear Algebra Review
- ▶ Quantum Mechanics
- ▶ Basics of Quantum Circuits
- ▶ Quantum Teleportation

Linear Algebra Review

Warning

This would be very hard to follow without prior experience in linear algebra. For reviews of linear algebra, look at Chapter 2.1 of Nielsen and Chuang

Summary of Dirac Notation (Borrowed from Nielsen and Chuang)

Notation	Description
z^*	Complex conjugate of the complex number z . $(1 + i)^* = 1 - i$
$ \psi\rangle$	Vector. Also known as a <i>ket</i> .
$\langle\psi $	Vector dual to $ \psi\rangle$. Also known as a <i>bra</i> .
$\langle\varphi \psi\rangle$	Inner product between the vectors $ \varphi\rangle$ and $ \psi\rangle$.
$ \varphi\rangle \otimes \psi\rangle$	Tensor product of $ \varphi\rangle$ and $ \psi\rangle$.
$ \varphi\rangle \psi\rangle$	Abbreviated notation for tensor product of $ \varphi\rangle$ and $ \psi\rangle$.
A^*	Complex conjugate of the A matrix.
A^T	Transpose of the A matrix.
$A^\dagger$	Hermitian conjugate or adjoint of the A matrix, $A^\dagger = (A^T)^*$. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^\dagger = \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix}.$
$\langle\varphi A \psi\rangle$	Inner product between $ \varphi\rangle$ and $A \psi\rangle$. Equivalently, inner product between $A^\dagger \varphi\rangle$ and $ \psi\rangle$.

1. Vectors and Dirac notation

Let $\mathcal{H} \cong \mathbb{C}^N$ be a finite-dimensional inner-product space with orthonormal basis $\{|0\rangle, \dots, |N-1\rangle\}$.

$$|\psi\rangle = \sum_{j=0}^{N-1} \alpha_j |j\rangle \quad \longleftrightarrow \quad \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{N-1} \end{pmatrix}.$$

- ▶ A "ket" $|\psi\rangle$ is a column vector.
- ▶ The corresponding "bra" is the conjugate transpose:

$$\langle\psi| = |\psi\rangle^\dagger.$$

- ▶ Inner (braket) and outer (ketbra) products are

$$\langle\phi | \psi\rangle = \phi^\dagger \psi, \quad |\psi\rangle\langle\phi| = \psi\phi^\dagger.$$

2. Orthonormal bases and coordinates

For an orthonormal basis $\{|u_j\rangle\}_{j=0}^{N-1}$,

$$\langle u_j | u_k \rangle = \delta_{jk}, \quad \sum_j |u_j\rangle \langle u_j| = I.$$

Every vector has the expansion

$$|\psi\rangle = \sum_j \langle u_j | \psi \rangle |u_j\rangle.$$

- ▶ The coefficients are projections onto the basis vectors.
- ▶ Changing basis changes the coordinates, not the vector.
- ▶ If the columns of U are $|u_0\rangle, \dots, |u_{N-1}\rangle$, then U is unitary and $U^\dagger |\psi\rangle$ gives the coordinates in this basis.

3. Operator Zoo

A linear operator is represented by a matrix $A \in \mathbb{C}^{N \times N}$.

Operator	Condition	Role
Hermitian	$A = A^\dagger$	observable / generator
Unitary	$U^\dagger U = UU^\dagger = I$	reversible evolution
Projector	$P = P^\dagger = P^2$	subspace / measurement event

- ▶ Unitaries preserve inner products and norms.
- ▶ A rank-one projector onto $\text{span}\{|u\rangle\}$ is $P = |u\rangle\langle u|$.
- ▶ A Hermitian operator has real eigenvalues and an orthonormal eigenbasis ("Spectral Theorem")

4. Tensor products

If systems A and B have state spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the joint space is

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

For vectors and operators,

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}, \quad (A \otimes B)(|u\rangle \otimes |v\rangle) = A|u\rangle \otimes B|v\rangle.$$

- ▶ Dimensions multiply: $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim \mathcal{H}_A \dim \mathcal{H}_B$.
- ▶ An n -qubit system has state space $(\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$.
- ▶ Not every vector in a tensor-product space is itself a tensor product. ¹

¹Quantum Entanglement is a consequence of this, covered later this lecture!

Quantum Mechanics

A quantum bit

A classical bit is either 0 or 1. A qubit has basis states²,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

It may also be in a superposition of its basis states

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The complex numbers α and β are called *amplitudes*.

²This is known as the "computational basis", from the rules of linear algebra you can obviously write this in any basis you want

Measuring a qubit

Measurements "collapse" the qubit into its basis states³

Outcome	Probability	State afterwards
0	$ \alpha ^2$	$ 0\rangle$
1	$ \beta ^2$	$ 1\rangle$

For example,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \Rightarrow \quad \Pr(0) = \Pr(1) = \frac{1}{2}.$$

³Again you can measure in any basis you choose but the computational basis here is chosen for convenience

Relative and Global Phase

Consider the two states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

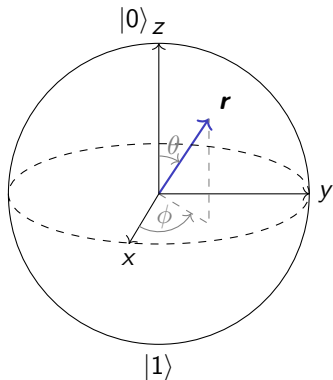
The minus sign above is a "relative phase". It has observable consequences when the amplitudes are later recombined.

However, multiplying the whole state by a global phase does not change the square of the amplitudes (as $|e^{i\phi}|^2 = 1$), so it does not change the measurement probabilities

$$e^{i\gamma}|\psi\rangle \sim |\psi\rangle.$$

These are the same quantum state

The Bloch sphere



After ignoring global phase, every pure qubit state can be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle.$$

The angles (θ, ϕ) specify a point on the sphere

$$\mathbf{r} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}, \quad \|\mathbf{r}\| = 1.$$

Quantum gates

A quantum gate is represented by a unitary matrix:

$$U^\dagger U = I.$$

Unitarity preserves inner products and normalization,

$$\langle U\phi | U\psi \rangle = \langle \phi | \psi \rangle, \quad \|U|\psi\rangle\| = \|\psi\|.$$

Linearity:

$$U(\alpha|0\rangle + \beta|1\rangle) = \alpha U|0\rangle + \beta U|1\rangle.$$

The X and Z gates

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle.$$

The X gate flips the basis states.

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

The Z gate changes relative phase.

The Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle.$$

It converts between the computational and Hadamard bases:

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle, \quad H^2 = I.$$

Interference in one qubit

Start in the state

$$|0\rangle \xrightarrow{H} |+\rangle.$$

Applying Z^b encodes b as a relative phase:

$$b = 0 : \quad |0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z^0} |+\rangle \xrightarrow{H} |0\rangle,$$

$$b = 1 : \quad |0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle.$$

Thus,

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle.$$

Relative phases can interfere and have observable effects!

Two qubits

Two qubits have four computational-basis states:

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

Their general state is

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle,$$

$$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1.$$

Measuring both qubits gives $x \in \{00, 01, 10, 11\}$ with probability $|\alpha_x|^2$.

Measuring one qubit

For

$$|\psi\rangle = \sum_{x,y \in \{0,1\}} \alpha_{xy} |xy\rangle,$$

measuring only the first qubit gives

$$\Pr(0) = |\alpha_{00}|^2 + |\alpha_{01}|^2, \quad \Pr(1) = |\alpha_{10}|^2 + |\alpha_{11}|^2.$$

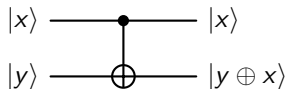
Conditioning on outcome 0, the state becomes

$$\frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}}.$$

The controlled-NOT gate

The controlled-NOT gate flips the second qubit exactly when the first qubit is 1:

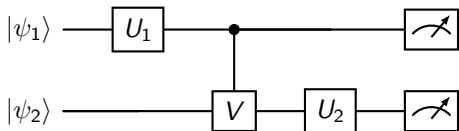
$$|x, y\rangle \mapsto |x, y \oplus x\rangle.$$



Its action on an arbitrary state follows by linearity. For example,

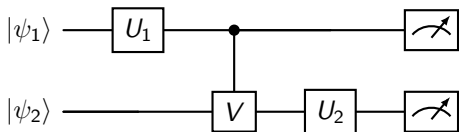
$$\frac{|00\rangle + |10\rangle}{\sqrt{2}} \mapsto \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Quantum circuits



- ▶ Each line represents a qubit; time runs from left to right.
- ▶ Gates on separate qubits act in parallel by tensor product.
- ▶ Successive gates compose by matrix multiplication.
- ▶ A controlled- V applies V to the target when the control is 1.

Quantum circuits



- ▶ Each line represents a qubit; time runs from left to right.
- ▶ Gates on separate qubits act in parallel by tensor product.
- ▶ Successive gates compose by matrix multiplication.
- ▶ A controlled- V applies V to the target when the control is 1.

$$(|\psi_1\rangle \otimes |\psi_2\rangle) \xrightarrow{U_1 \otimes I} (U_1 \otimes I)(|\psi_1\rangle \otimes |\psi_2\rangle) \xrightarrow{C_V} C_V(U_1 \otimes I)(|\psi_1\rangle \otimes |\psi_2\rangle) \xrightarrow{I \otimes U_2} (I \otimes U_2)C_V(U_1 \otimes I)(|\psi_1\rangle \otimes |\psi_2\rangle)$$

No-cloning theorem: proof

Suppose one unitary copied two states using a fixed blank state $|s\rangle$:

$$U|\psi_1\rangle|s\rangle = |\psi_1\rangle|\psi_1\rangle, \quad U|\psi_2\rangle|s\rangle = |\psi_2\rangle|\psi_2\rangle.$$

Since a unitary preserves inner products,

$$\langle\psi_1 | \psi_2\rangle = \langle\psi_1 | \psi_2\rangle^2.$$

Hence $\langle\psi_1 | \psi_2\rangle$ must be 0 or 1. A single unitary therefore cannot copy arbitrary nonorthogonal states.

Product states and entangled states

A two-qubit product state has the form

$$|\psi\rangle_A \otimes |\phi\rangle_B.$$

For example,

$$|+\rangle|0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

cannot be written as a product state.

DIY: Try to prove this

Product states and entangled states

Suppose it can be written as a product state:

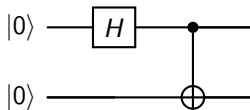
$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

comparison of coefficients would require

$$ac = bd = \frac{1}{\sqrt{2}}, \quad ad = bc = 0,$$

So $1/2 = abcd = 0$, contradiction!

Creating a Bell pair



By linearity,

$$|00\rangle \xrightarrow{H \otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Quantum teleportation

Quantum teleportation: Setup

Alice has an unknown qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

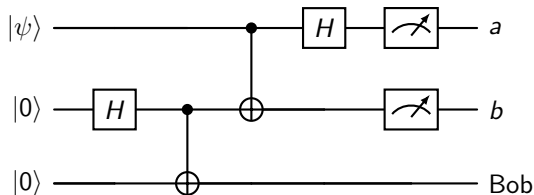
and wants to send it to Bob

Alice and Bob are given

- ▶ one shared Bell pair $(|00\rangle + |11\rangle)/\sqrt{2}$;
- ▶ a classical channel from Alice to Bob.

Is there a way Alice can send her qubit to Bob only using this?

Teleportation circuit



1. Prepare the lower two qubits as a Bell pair.
2. Alice applies CNOT and H , then measures her qubits.
3. Alice sends the two outcomes a, b to Bob.
4. Bob applies X^b followed by Z^a .

DIY: Why does this teleport the state?

Why teleportation works

After Alice's CNOT and Hadamard gates, the joint state is

$$\frac{1}{2}(|00\rangle|\psi\rangle + |01\rangle X|\psi\rangle + |10\rangle Z|\psi\rangle + |11\rangle XZ|\psi\rangle).$$

Alice's outcome ab	Bob's state	Bob applies
00	$ \psi\rangle$	I
01	$X \psi\rangle$	X
10	$Z \psi\rangle$	Z
11	$XZ \psi\rangle$	ZX

- ▶ Alice obtains each $ab \in \{00, 01, 10, 11\}$ with probability $1/4$.
- ▶ Bob can get $|\psi\rangle$ just by applying the inverse Pauli Operators on his own states based on classical bits a, b
- ▶ No violation of No Cloning Theorem!

Summary

Topics Covered:

- ▶ Linear Algebra
- ▶ Quantum Mechanics and Circuits
- ▶ Quantum Teleportation

Reading:

- ▶ Nielsen and Chuang 1.1-1.3
- ▶ de Wolf Chapter 1

For a more rigorous introduction to Linear Algebra and Quantum Mechanics: Nielsen and Chuang 2.1-2.2

Next Lecture

To be covered:

- ▶ More Quantum Mechanics!
- ▶ Deutsch Josza Algorithm
- ▶ Bernstein Vazirani Algorithm

Pre-Reading:

- ▶ Nielsen and Chuang 1.4.1 - 1.4.5 (Skip Complexity Part until Ian's Lecture)
- ▶ de Wolf Chapter 2 (Skip 2.1.1)